2) use lump model

$$A = N 4\pi R^2 = \frac{m}{\rho_s \frac{4}{3}\pi R^3} 4\pi r^2 = \frac{3m}{\rho_s R} = 0.11\ m^2$$

$$m_s c_{ps} \frac{dT_s}{dt} = -hA(T_s - T_\infty)$$

$$T_\infty = 20^\circ C$$

$$\frac{dT_s}{(T_s - T_\infty)} = \frac{-hA}{m_s c_{ps}} dt$$

$$\ln\left(\frac{T_{sf} - T_\infty}{T_{si} - T_\infty}\right) = t \frac{-hA}{m_s c_{ps}}$$

$$t = \frac{m_s c_{ps}}{-hA} \ln\left(\frac{T_{sf} - T_\infty}{T_{si} - T_\infty}\right)$$

$$t = \frac{900}{500(0.111)} \ln\left(\frac{40-20}{90-20}\right) = 20\ s$$

2) Say

$$m_w c_{pw} \frac{dT_w}{dt} = hA(T_s - T_w)$$

$$C_w \qquad UA$$

Say

$$\frac{d}{dt}(C_s T_s + C_w T_w) = 0$$

$$\frac{dT_s}{dT_w} = -\frac{C_w}{C_s}$$

$$\int dT_s = \int -\frac{C_w}{C_s} dT_w$$

$\frac{\ }{500(0.111)}(90-30)$

2) Say

$m_w c_{pw} \frac{dT_w}{dt} = hA(T_s - T_w)$

c_w $\quad$ UA

Say

$\frac{d}{dt}(c_sT_s + c_wT_w) = 0$

b/c conservation of energy

$c_sT_s + c_wT_w = M$ constant

$T_w = \frac{M - c_sT_s}{c_w}$

$\frac{dT_s}{dT_w} = -\frac{c_w}{c_s}$

$\int dT_s = \int -\frac{c_w}{c_s}\, dT_w$

$T_{s1} - T_f = -\frac{c_w}{c_s}(T_f - T_{w0})$

$c_s\dot{T}_s = -hA(T_s - T_w)$

$T_w = \frac{M - c_sT_s}{c_w}$

$= -hA\left(T_s - \frac{M}{c_w} + \frac{c_sT_s}{c_w}\right)$

$c_s\dot{T}_s = -hA\left(T_s\left(1 + \frac{c_s}{c_w}\right) - \frac{M}{c_w}\right)$

$\dot{T}_s + \frac{hA}{c_s}\left(1 + \frac{c_s}{c_w}\right) = \frac{hA}{c_s}\frac{M}{c_w}$

Say

$c_s(T_{si} - T_{sf}) = c_w(T_f - T_{wi})$

$$T_f = \frac{c_s T_{si} + c_w T_{wi}}{c_s + c_w}$$

$$T_s(t) = T_f + (T_{si} - T_f)e^{-t/h}$$

plug and chug to

get T_s

$t = 275$

1) $m_c c_{pc} = 5{,}000$

$\Delta T_c = 210 - 90$

$q = \Delta T_c m_c c_{pc} = 6\times10^5$ BTU/h

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$m_c C_{pc} = 5{,}000$

$\Delta T_c = 210 - 90$

$q = \Delta T_c m_c C_{pc} = 6\times10^5$ BTU/h

$q = m_H C_{pH} \Delta T_H$ $\quad \Delta T_H = 260 - X$) set equal to 3et $\quad X = 200\,°F$

$q = UA \dfrac{\Delta T_A - \Delta T_B}{\ln(\Delta T_A / \Delta T_B)}$ $\quad \Delta T_A = T_{h\,in} - T_{c\,out}$

$A = 2\pi r L \cdot N_t$ $\quad \Delta T_B = T_{h\,out} - T_{c\,in}$

$q = U_o 2\pi r L \cdot N_t \Delta T_{lm}$

Solve for L

$L = \dfrac{2009}{U_o}$ ft ?

if $U_o = 150$

$L = 13.4$ ft